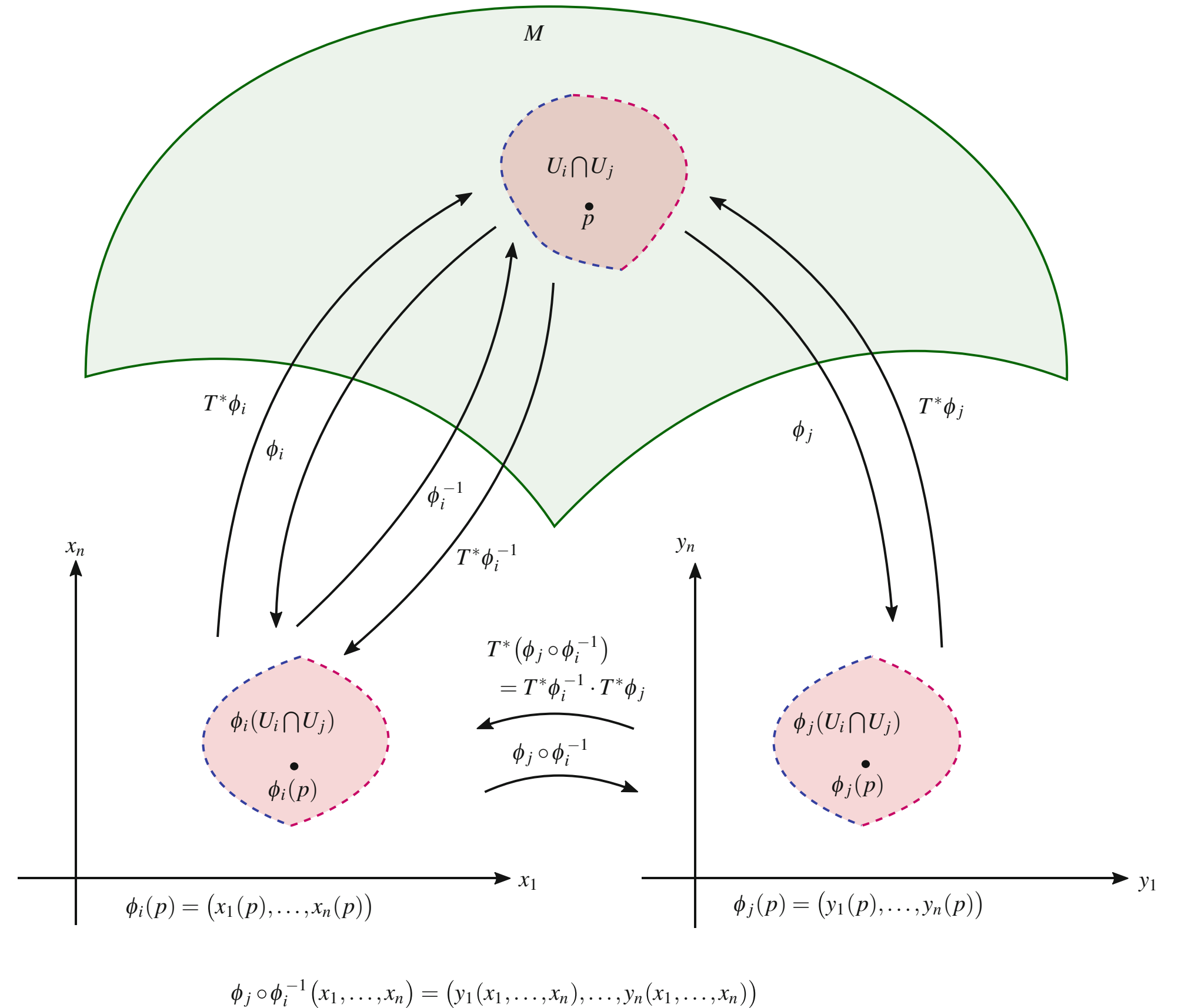

M
$U_i \cap U_j$
p
$T^*\phi_i$
ϕ_i
ϕ_i^{-1}
$T^*\phi_i^{-1}$
ϕ_j
$T^*\phi_j$
x_n
y_n
$T^*\left(\phi_j \circ \phi_i^{-1}\right)$
$= T^*\phi_i^{-1} \cdot T^*\phi_j$
$\phi_j \circ \phi_i^{-1}$
$\phi_i(U_i \cap U_j)$
$\phi_i(p)$
$\phi_j(U_i \cap U_j)$
$\phi_j(p)$
x_1
y_1
$\phi_i(p) = \left(x_1(p), \ldots, x_n(p)\right)$
$\phi_j(p) = \left(y_1(p), \ldots, y_n(p)\right)$
$\phi_j \circ \phi_i^{-1}\left(x_1, \ldots, x_n\right) = \left(y_1(x_1, \ldots, x_n), \ldots, y_n(x_1, \ldots, x_n)\right)$